

Laplace Transform Theory

1 Introduction

Let

$$\Psi_b(t) = \Psi_{b0} \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}. \quad (1)$$

It follows that

$$\bar{\Psi}_b(g) = \Psi_{b0} \int_0^\infty e^{-gt} dt = \frac{\Psi_{b0}}{g}. \quad (2)$$

Now, the Laplace transformed reconnected magnetic flux is

$$\bar{\Psi}_0(g) = \frac{E_{sb} F_s(\hat{g}) \bar{\Psi}_b(g)}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}} = E_{sb} \Psi_{b0} \frac{1}{g} \frac{F_s(\hat{g})}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}, \quad (3)$$

where $\hat{g} = S^{1/3} g \tau_H$. The reconnected flux is

$$\Psi_0(t) = \frac{1}{2\pi i} \int_c \bar{\Psi}_0(g) e^{gt} dg, \quad (4)$$

where c is the Bromwich contour. Thus, we obtain

$$\Psi_0(\hat{t}) = \frac{1}{2\pi i} \int_c \bar{\Psi}_0(\hat{g}) e^{\hat{g}\hat{t}} d\hat{g}, \quad (5)$$

where $\hat{t} = t/(S^{1/3} \tau_H)$ and

$$\bar{\Psi}_0(\hat{g}) = E_{sb} \Psi_{b0} \frac{1}{\hat{g}} \frac{F_s(\hat{g})}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}. \quad (6)$$

2 Non-Constant- ψ Regimes

We shall assume that $|S^{1/3} \hat{\Delta}_s| \gg |E_{ss}|$ in the non-constant- ψ regimes, so

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{E_{sb} \Psi_{b0}}{S^{1/3}} \frac{1}{\hat{g}} \frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})}. \quad (7)$$

2.1 Inertial Regime

In the inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}, \quad (8)$$

$$F_s(\hat{g}) = -\frac{2}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E) [\hat{g} + i (Q_E + Q_i)]}, \quad (9)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi} \frac{1}{[\hat{g} + i (Q_E + Q_e)] (\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}} \quad (10)$$

Thus,

$$\begin{aligned} \bar{\Psi}_0(\hat{g}) &= \frac{\Psi_\infty \epsilon}{i (Q_E + Q_e)} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i (Q_E + Q_e)} \right] \\ &\times \frac{1}{(\hat{g} + i Q_E)^{1/2} [\hat{g} + i (Q_E + Q_i)]^{1/2}}, \end{aligned} \quad (11)$$

where

$$\epsilon = \frac{2}{\pi} \frac{(-E_{ss})}{S^{1/3}}, \quad (12)$$

$$\Psi_\infty = \frac{E_{sb} \Psi_{b0}}{(-E_{ss})}. \quad (13)$$

Under normal circumstances $0 < \epsilon \ll 1$. Moreover, Ψ_∞ is the final amount of driven reconnection in the absence of suppression.

Now, if

$$\bar{\Psi}_0(\hat{g}) = \frac{1}{\hat{g} + a} \bar{H}(\hat{g}) \quad (14)$$

then (ERD:¹ 4.1.20)

$$\Psi_0(\hat{t}) = \int_0^{\hat{t}} e^{a(\hat{t}' - \hat{t})} H(\hat{t}') d\hat{t}'. \quad (15)$$

¹Erdélyi, et al. *Tables of Integral Transforms*, vol. I.

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + \mathrm{i} Q_E)^{1/2} [\hat{g} + \mathrm{i} (Q_E + Q_i)]^{1/2}} \quad (16)$$

$$= \frac{1}{(\hat{g}^2 + \alpha \hat{g} + \beta)^{1/2}}, \quad (17)$$

where

$$\alpha = \mathrm{i} (2 Q_E + Q_i), \quad (18)$$

$$\beta = -Q_E (Q_E + Q_i). \quad (19)$$

Note that

$$\beta - \frac{1}{4} \alpha^2 = -Q_E (Q_E + Q_i) + \frac{1}{4} (2 Q_E + Q_i)^2 = \frac{Q_i^2}{4}. \quad (20)$$

However (ERD: 5.3.34),

$$H(\hat{t}) = \mathrm{e}^{-\alpha \hat{t}/2} J_0[(\beta - \alpha^2/4)^{1/2} \hat{t}] \quad (21)$$

$$= \mathrm{e}^{-\mathrm{i} (Q_E + Q_i/2) \hat{t}} J_0\left(\frac{Q_i}{2} \hat{t}\right). \quad (22)$$

Thus,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{\Psi_\infty \epsilon}{\mathrm{i} (Q_E + Q_e)} \int_0^{\hat{t}} \left[1 - \mathrm{e}^{\mathrm{i} (Q_E + Q_e) (\hat{t}' - \hat{t})} \right] \\ &\quad \times \mathrm{e}^{-\mathrm{i} (Q_E + Q_i/2) \hat{t}'} J_0\left(\frac{Q_i}{2} \hat{t}'\right) d\hat{t}'. \end{aligned} \quad (23)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}^2}{\Gamma(3)}. \quad (24)$$

At large $\hat{t}$,

$$\begin{aligned} \Psi_0(\hat{t}) &\simeq \frac{\Psi_\infty \epsilon}{\mathrm{i} (Q_E + Q_e)} \int_0^\infty \left[1 - \mathrm{e}^{\mathrm{i} (Q_E + Q_e) (\hat{t}' - \hat{t})} \right] \\ &\quad \times \mathrm{e}^{-\mathrm{i} (Q_E + Q_i/2) \hat{t}'} J_0\left(\frac{Q_i}{2} \hat{t}'\right) d\hat{t}'. \end{aligned} \quad (25)$$

But (GR:² 6.621.4)

$$\int_0^\infty e^{-\alpha x} J_0(\beta x) dx = \frac{1}{\sqrt{\alpha^2 + \beta^2}}, \quad (26)$$

so

$$\Psi_{b0}(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{i(Q_E + Q_i)} \left\{ \frac{1}{(iQ_E)^{1/2} [i(Q_E + Q_i)]^{1/2}} - \frac{e^{-i(Q_E + Q_e)\hat{t}}}{(iQ_e)^{1/2} [i(Q_e - Q_i)]^{1/2}} \right\}. \quad (27)$$

2.2 Viscous-Inertial Regime

In the Viscous-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{1/4} P_\varphi^{1/4}}, \quad (28)$$

$$F_s(\hat{g}) = -\frac{\pi^{1/2}}{2} \frac{\Gamma(1/4)}{\Gamma(3/4)} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2} P_\varphi^{1/2}}, \quad (29)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{1}{\pi^{1/2}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4} P_\varphi^{1/4}}. \quad (30)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}, \quad (31)$$

where

$$\epsilon = \frac{(-E_{ss})}{\pi^{1/2} S^{1/3} P_\varphi^{1/4}}. \quad (32)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{5/4}}, \quad (33)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(5/4)} \hat{t}^{1/4} e^{-i(Q_E + Q_e)\hat{t}}. \quad (34)$$

²I.S. Gradshteyn and I.M. Ryzhik, *Table of Integrals, Series, and Products*, Corrected and Enlarged Edition

Hence,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{\Gamma(5/4)} \int_0^{\hat{t}} \hat{t}'^{1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (35)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}^{5/4}}{\Gamma(9/4)}. \quad (36)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{\Gamma(5/4)} \int_0^\infty \hat{t}'^{1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (37)$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} e^{-iqx} dx = \frac{\Gamma(\nu)}{(iq)^\nu}, \quad (38)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{[i(Q_E + Q_e)]^{5/4}}. \quad (39)$$

2.3 Diffusive-Inertial Regime

In the Diffusive-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = -\frac{\pi D}{[\hat{g} + i(Q_E + Q_e)]^{1/2} \iota_e^{1/2} P_\perp^{1/2}}, \quad (40)$$

$$F_s(\hat{g}) = -\frac{2D^2}{[\hat{g} + i(Q_E + Q_e)]^2 \iota_e P_\perp}, \quad (41)$$

$$\frac{F_s(\hat{g})}{\hat{\Delta}_s(\hat{g})} = \frac{2}{\pi} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2} \iota_e^{1/2} P_\perp^{1/2} D^{-1}}. \quad (42)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2}}, \quad (43)$$

where

$$\epsilon = \frac{2}{\pi} \frac{(-E_{ss})}{S^{1/3} \iota_e^{1/2} P_\perp^{1/2} D^{-1}}. \quad (44)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/2}}, \quad (45)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(3/2)} \hat{t}^{1/2} e^{-i(Q_E + Q_e)\hat{t}}. \quad (46)$$

Hence,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{\Gamma(3/2)} \int_0^{\hat{t}} \hat{t}'^{1/2} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (47)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}^{3/2}}{\Gamma(5/2)}. \quad (48)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{\Gamma(3/2)} \int_0^\infty \hat{t}'^{1/2} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \quad (49)$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} e^{-iqx} dx = \frac{\Gamma(\nu)}{(iq)^\nu}, \quad (50)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{[i(Q_E + Q_e)]^{3/2}}. \quad (51)$$

3 Constant- ψ Regimes

We shall not necessarily assume that $|S^{1/3} \hat{\Delta}_s| \gg |E_{ss}|$ in the constant- ψ regimes, which are characterized by $F_s(\hat{g}) = 1$, so

$$\bar{\Psi}_0(\hat{g}) \simeq E_{sb} \Psi_{b0} \frac{1}{\hat{g}} \frac{1}{S^{1/3} \hat{\Delta}_s(\hat{g}) - E_{ss}}. \quad (52)$$

3.1 Viscous-Resistive Regime

In the Viscous-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{6^{2/3} \pi \Gamma(5/6)}{\Gamma(1/6)} [\hat{g} + i(Q_E + Q_e)] P_\varphi^{1/6}, \quad (53)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{\epsilon}{[\hat{g} + i(Q_E + Q_e)] + \epsilon}, \quad (54)$$

where

$$\epsilon = \frac{\Gamma(1/6)}{6^{3/2} \pi \Gamma(5/6)} \frac{(-E_{ss})}{S^{1/3} P_\varphi^{1/6}}. \quad (55)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon} \left(\frac{1}{\hat{g}} - \frac{1}{\hat{g} + [i(Q_E + Q_e) + \epsilon]} \right), \quad (56)$$

giving

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon} \left(1 - e^{-[i(Q_E + Q_e) + \epsilon] \hat{t}} \right). \quad (57)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}}{\Gamma(2)}. \quad (58)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon}. \quad (59)$$

3.2 Diffusive-Resistive Regime

In the Diffusive-Resistive regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [\hat{g} + i(Q_E + Q_e)] \iota_e^{1/4} P_\perp^{1/4} D^{-1/2}, \quad (60)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)] + \epsilon}, \quad (61)$$

where

$$\epsilon = \frac{\Gamma(1/4)}{2\pi \Gamma(3/4)} \frac{(-E_{ss})}{S^{1/3} \iota_e^{1/4} P_\perp^{1/4} D^{-1/2}}. \quad (62)$$

Thus,

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon} \left(\frac{1}{\hat{g}} - \frac{1}{\hat{g} + [i(Q_E + Q_e) + \epsilon]} \right), \quad (63)$$

giving

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon} \left(1 - e^{-[i(Q_E + Q_e) + \epsilon] \hat{t}} \right). \quad (64)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}}{\Gamma(2)}. \quad (65)$$

At large $\hat{t}$,

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e) + \epsilon}. \quad (66)$$

3.3 Semi-Collisional Regime

In the Semi-Collisional regime,

$$\hat{\Delta}_s(\hat{g}) = \pi [\hat{g} + i(Q_E + Q_e)] (\hat{g} + iQ_E)^{1/2} D^{-1}, \quad (67)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)] (\hat{g} + iQ_E)^{1/2} + \epsilon}, \quad (68)$$

where

$$\epsilon = \frac{1}{\pi} \frac{(-E_{ss})}{S^{1/3} D^{-1}}. \quad (69)$$

If we neglect ϵ in the denominator then

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)] (\hat{g} + iQ_E)^{1/2}}, \quad (70)$$

giving

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{\Psi_\infty \epsilon}{i(Q_E + Q_e)} \left[\frac{1}{\hat{g}} - \frac{1}{\hat{g} + i(Q_E + Q_e)} \right] \frac{1}{(\hat{g} + iQ_E)^{1/2}}. \quad (71)$$

In this case,

$$\bar{H}(\hat{g}) = \frac{1}{(\hat{g} + iQ_E)^{1/2}}, \quad (72)$$

so (ERD: 5.4.1)

$$H(\hat{t}) = \frac{1}{\Gamma(1/2)} \hat{t}^{-1/2} e^{-i Q_E \hat{t}}. \quad (73)$$

Hence,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{\Psi_\infty \epsilon}{\Gamma(1/2) i (Q_E + Q_e)} \\ &\times \int_0^{\hat{t}} \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})} \right] \hat{t}'^{-1/2} e^{-i Q_E \hat{t}'} d\hat{t}'. \end{aligned} \quad (74)$$

At small $\hat{t}$,

$$\Psi_0(\hat{t}) = \Psi_\infty \epsilon \frac{\hat{t}^{3/2}}{\Gamma(5/2)}. \quad (75)$$

At large $\hat{t}$,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{\Psi_\infty \epsilon}{\Gamma(1/2) i (Q_E + Q_e)} \\ &\times \int_0^\infty \left[1 - e^{i(Q_E + Q_e)(\hat{t}' - \hat{t})} \right] \hat{t}'^{-1/2} e^{-i Q_E \hat{t}'} d\hat{t}'. \end{aligned} \quad (76)$$

But (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} e^{-i q x} dx = \frac{\Gamma(\nu)}{(i q)^\nu}, \quad (77)$$

It follows that

$$\Psi_0(\hat{t}) = \frac{\Psi_\infty \epsilon}{i (Q_E + Q_e)} \left[\frac{1}{(i Q_E)^{1/2}} - \frac{e^{-i(Q_E + Q_e)\hat{t}}}{(-i Q_e)^{1/2}} \right]. \quad (78)$$

3.4 Resistive-Inertial Regime

In the Resistive-Inertial regime,

$$\hat{\Delta}_s(\hat{g}) = \frac{2\pi \Gamma(3/4)}{\Gamma(1/4)} [\hat{g} + i(Q_E + Q_e)]^{3/4} (\hat{g} + i Q_E)^{1/4} [\hat{g} + i(Q_E + Q_i)]^{1/4}, \quad (79)$$

so

$$\bar{\Psi}_0(\hat{g}) = \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/4} (\hat{g} + i Q_E)^{1/4} [\hat{g} + i(Q_E + Q_i)]^{1/4} + \epsilon}, \quad (80)$$

where

$$\epsilon = \frac{\Gamma(1/4)}{2\pi \Gamma(3/4)} \frac{(-E_{ss})}{S^{1/3}}. \quad (81)$$

If we neglect ϵ in the denominator then

$$\bar{\Psi}_0(\hat{g}) \simeq \frac{\Psi_\infty \epsilon}{\hat{g}} \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/4} (\hat{g} + iQ_E)^{1/4} [\hat{g} + i(Q_E + Q_i)]^{1/4}}. \quad (82)$$

Now, if

$$\bar{\Psi}_0(\hat{g}) \simeq \Psi_\infty \epsilon \frac{H(\hat{g})}{\hat{g}} \quad (83)$$

then

$$\Psi_0(\hat{t}) = \Psi_\infty \epsilon \int_0^{\hat{t}} H(\hat{t}') d\hat{t}'. \quad (84)$$

Furthermore, if

$$\bar{H}(\hat{g}) = \bar{H}_1(\hat{g}) \bar{H}_2(\hat{g}) \quad (85)$$

then

$$H(\hat{t}) = \int_0^{\hat{t}} H_1(\hat{t}') H_2(\hat{t} - \hat{t}') d\hat{t}'. \quad (86)$$

It follows that

$$\Psi_0(\hat{t}) = \Psi_\infty \epsilon \int_0^{\hat{t}} \left[\int_0^{\hat{t}'} H_1(\hat{t}'') H_2(\hat{t}' - \hat{t}'') d\hat{t}'' \right] d\hat{t}'. \quad (87)$$

In the present case

$$\bar{H}_1(\hat{g}) = \frac{1}{(\hat{g} + iQ_E)^{1/4} [\hat{g} + i(Q_E + Q_i)]^{1/4}}, \quad (88)$$

$$\bar{H}_2(\hat{g}) = \frac{1}{[\hat{g} + i(Q_E + Q_e)]^{3/4}}, \quad (89)$$

so (ERD: 5.4.5, AS:³ 9.6.3, ERD: 5.4.1)

$$H_1(\hat{t}) = \frac{\pi^{1/2}}{\Gamma(1/4)} Q_i^{1/4} \hat{t}^{-1/4} e^{-i(Q_E + Q_i/2)\hat{t}} J_{-1/4} \left(\frac{Q_i}{2} \hat{t} \right), \quad (90)$$

³M. Abramowitz and I.A. Stegun, *Handbook of Mathematical Functions* (Dover, New York NY, 1964).

$$H_2(\hat{t}) = \frac{1}{\Gamma(3/4)} \hat{t}^{-1/4} e^{-i(Q_E+Q_e)\hat{t}}. \quad (91)$$

Note that

$$\Gamma(1/4)\Gamma(3/4) = \sqrt{2}\pi. \quad (92)$$

Thus,

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{\Psi_\infty \epsilon Q_i^{1/4}}{\sqrt{2\pi}} \int_0^{\hat{t}} e^{-i(Q_E+Q_e)\hat{t}'} \left[\int_0^{\hat{t}'} (\hat{t}' - \hat{t}'')^{-1/4} \hat{t}''^{-1/4} \right. \\ &\quad \left. \times e^{i(Q_e-Q_i/2)\hat{t}''} J_{-1/4}\left(\frac{Q_i}{2}\hat{t}''\right) d\hat{t}'' \right] d\hat{t}'. \end{aligned} \quad (93)$$

At small $\hat{t}$ (AS: 9.1.10),

$$J_{-1/4}\left(\frac{Q_i}{2}\hat{t}\right) \simeq \frac{\sqrt{2}Q_i^{-1/4}\hat{t}^{-1/4}}{\Gamma(3/4)}, \quad (94)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{\sqrt{\pi}\Gamma(3/4)} \int_0^{\hat{t}} \left[\int_0^{\hat{t}'} (\hat{t}' - \hat{t}'')^{-1/4} \hat{t}''^{-1/2} d\hat{t}'' \right] d\hat{t}'. \quad (95)$$

But,

$$\int_0^{\hat{t}'} (\hat{t}' - \hat{t}'')^{-1/4} \hat{t}''^{-1/2} d\hat{t}'' = \hat{t}'^{1/4} \int_0^1 (1-x)^{-1/4} x^{-1/2} dx, \quad (96)$$

and (AS: 6.2.1, 6.2.2)

$$\int_0^1 (1-x)^{-1/4} x^{-1/2} dx = \frac{\Gamma(1/2)\Gamma(3/4)}{\Gamma(5/4)}, \quad (97)$$

so

$$\Psi_0(\hat{t}) \simeq \Psi_\infty \epsilon \frac{\hat{t}^{5/4}}{\Gamma(9/4)}. \quad (98)$$

At large $\hat{t}$,

$$\begin{aligned} \Psi_0(\hat{t}) &\simeq \frac{\Psi_\infty \epsilon Q_i^{1/4}}{\sqrt{2\pi}} \int_0^\infty e^{-i(Q_E+Q_e)\hat{t}'} \left[\int_0^{\hat{t}'} (\hat{t}' - \hat{t}'')^{-1/4} \hat{t}''^{-1/4} \right. \\ &\quad \left. \times e^{i(Q_e-Q_i/2)\hat{t}''} J_{-1/4}\left(\frac{Q_i}{2}\hat{t}''\right) d\hat{t}'' \right] d\hat{t}', \end{aligned} \quad (99)$$

so

$$\begin{aligned} \Psi_0(\hat{t}) &\simeq \frac{\Psi_\infty \epsilon Q_i^{1/4}}{\sqrt{2\pi}} \int_0^\infty \left[\int_{\hat{t}''}^\infty (\hat{t}' - \hat{t}'')^{-1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' \right] \\ &\times \hat{t}''^{-1/4} e^{i(Q_e - Q_i/2)\hat{t}''} J_{-1/4}\left(\frac{Q_i}{2}\hat{t}''\right) d\hat{t}''. \end{aligned} \quad (100)$$

But,

$$\int_{\hat{t}''}^\infty (\hat{t}' - \hat{t}'')^{-1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' = e^{-i(Q_E + Q_e)\hat{t}''} \int_0^\infty x^{-1/4} e^{-i(Q_E + Q_e)x} dx, \quad (101)$$

and (GR: 3.381.5),

$$\int_0^\infty x^{\nu-1} e^{-iqx} dx = \frac{\Gamma(\nu)}{(iq)^\nu}, \quad (102)$$

so

$$\int_{\hat{t}''}^\infty (\hat{t}' - \hat{t}'')^{-1/4} e^{-i(Q_E + Q_e)\hat{t}'} d\hat{t}' = e^{-i(Q_E + Q_e)\hat{t}''} \frac{\Gamma(3/4)}{[i(Q_E + Q_e)]^{3/4}}. \quad (103)$$

It follows that

$$\begin{aligned} \Psi_0(\hat{t}) &= \frac{\Psi_\infty \epsilon Q_i^{1/4}}{\sqrt{2\pi}} \frac{\Gamma(3/4)}{[i(Q_E + Q_e)]^{3/4}} \\ &\times \int_0^\infty \hat{t}''^{-1/4} e^{-i(Q_E + Q_i/2)\hat{t}''} J_{-1/4}\left(\frac{Q_i}{2}\hat{t}''\right) d\hat{t}''. \end{aligned} \quad (104)$$

But (GR: 6.623.1),

$$\begin{aligned} &\int_0^\infty \hat{t}''^{-1/4} e^{-i(Q_E + Q_i/2)\hat{t}''} J_{-1/4}\left(\frac{Q_i}{2}\hat{t}''\right) d\hat{t}'' \\ &= \frac{(Q_i)^{-1/4} \Gamma(1/4)}{\sqrt{\pi} (iQ_E)^{1/4} [i(Q_E + Q_i)]^{1/4}}. \end{aligned} \quad (105)$$

Now (AS: 6.1.17),

$$\Gamma(3/4) \Gamma(1/4) = \sqrt{2} \pi, \quad (106)$$

so

$$\Psi_0(\hat{t}) \simeq \frac{\Psi_\infty \epsilon}{[i(Q_E + Q_e)]^{3/4} (iQ_E)^{1/4} [i(Q_E + Q_i)]^{1/4}}. \quad (107)$$